



# Croatian Open Competition in Informatics

Round 1, October 25<sup>th</sup> 2025

## Tasks

| Task             | Time limit  | Memory limit | Score |
|------------------|-------------|--------------|-------|
| <b>Rima</b>      | 1 second    | 512 MiB      | 50    |
| <b>Krugomet</b>  | 1 second    | 512 MiB      | 70    |
| <b>Harmonija</b> | 2.5 seconds | 1024 MiB     | 110   |
| <b>Kraljica</b>  | 5 seconds   | 512 MiB      | 110   |
| <b>Zagi</b>      | 3 seconds   | 512 MiB      | 110   |
| <b>Total</b>     |             |              | 450   |



## Task Rima

In Croatian class, Jakov reads a poem consisting of  $n$  lines and  $n/4$  stanzas. Lines 1, 2, 3, and 4 make up the first stanza, lines 5, 6, 7, and 8 the second stanza, and so on. Each line in each stanza consists of exactly  $m$  words.

We say that two lines rhyme if the last  $k$  letters of their last words match. If the last word in one of the lines consists of fewer than  $k$  letters, then the lines do not rhyme.

At the stanza level, we distinguish three types of rhyme schemes:

- AABB – lines 1 and 2 rhyme, and lines 3 and 4 rhyme
- ABAB – lines 1 and 3 rhyme, and lines 2 and 4 rhyme
- ABBA – lines 1 and 4 rhyme, and lines 2 and 3 rhyme

The teacher asked Jakov to determine in how many stanzas of the poem each of these three types of rhyme schemes appears. Help Jakov answer the teacher correctly and get an A in Croatian class!

### Input

The first line contains the natural numbers  $n$ ,  $m$  and  $k$  ( $1 \leq n \leq 500$  and  $n$  is divisible by 4,  $1 \leq m, k \leq 20$ ) from the text of the task.

The next  $n$  lines contain  $m$  words each, with each word being made up of at most 20 lowercase letters of the English alphabet.

### Output

In the first and only line, write three numbers: the number of stanzas in which AABB rhyme appears, the number of stanzas in which ABAB rhyme appears, and the number of stanzas in which ABBA rhyme appears.

### Scoring

| Subtask | Points | Constraints                          |
|---------|--------|--------------------------------------|
| 1       | 15     | $n = 4$                              |
| 2       | 15     | Each word is made up of 1 character. |
| 3       | 20     | No additional constraints.           |





## Examples

input

```
8 2 3
aa bb
cc bb
dd ee
ff ee
auu aaaaaahh
auu wer
asdf power
lol kldahh
```

output

```
0 0 1
```

input

```
8 2 2
ja programiram
mjesec listopad
ponekad chillam
voda vodopad
banana jabuka
fiziku znam
teska odluka
njam njam
```

output

```
0 2 0
```

input

```
4 4 2
pas konj zec macka
trokut teziste poluravnina tocka
nogomet tenis ragbi odbojka
sir mlijeko kulen sunka
```

output

```
1 1 1
```

**Clarification of the first example:** In the first stanza, not a single pair of lines can rhyme, because each last word has only 2 letters, and  $k = 3$ . The second stanza follows an ABBA rhyme scheme because the 1st and 4th lines rhyme (aaaaaahh and kldahh), and the 2nd and 3rd lines also rhyme (wer and power). Thus, AABB and ABAB rhymes occur 0 times, while ABBA rhyme occurs once.

**Clarification of the third example:** In the only stanza of this poem, the last two letters of the last word in each line are *ka*, so all lines rhyme with each other. It follows that all three types of rhyme schemes appear in this stanza: AABB, ABAB and ABBA.



## Task Krugomet

$n$  students gathered in the schoolyard one sunny day and stood in a large circle. Each person had his own box of balls (person  $i$  has  $a_i$  balls), as well as his crush (maybe even himself), denoted by  $s_i$ .

They decided to play their favorite game, krugomet. Playing krugomet consists of  $k$  rounds. In each round, all participants in the game **simultaneously** throw all their balls towards their crush. The students are skilled and agile, so they will always catch all the balls thrown to them in a round. After the balls are caught, everyone gets ready for the next round, and the whole process is repeated.



The teacher fell asleep on the bench and left the students to play alone. When he woke up, the students asked him to tell them who won, i.e. to answer 2 questions:

- How many balls does the student with the most balls have after  $k$  rounds of play?
- Who collected the most balls after  $k$  rounds of play?

The teacher knows how many students played krugomet  $n$ , the sequence of crushes  $s_i$ , the initial number of balls for each student  $a_i$ , and how many rounds of play  $k$  were played. Unfortunately, the teacher doesn't have time to count how many balls each person has, so he leaves that up to you.

Help him and determine the list of students who collected the most balls. In case there is more than one person with the most balls, print their indices in order from **smaller** to **larger**.

### Input

In the first line are the two natural numbers  $n$  and  $k$  ( $1 \leq n \leq 10^5$ ,  $1 \leq k \leq 10^9$ ) from the text of the task.

In the second line, there are  $n$  natural numbers  $a_i$  ( $1 \leq a_i \leq 1000$ ), numbers from the text of the task

In the third line are  $n$  natural numbers  $s_i$  ( $1 \leq s_i \leq n$ ), numbers from the text of the task.

### Output

In the first line, print a single number - the number of balls collected by the student with the most balls. In the second line, print the indices of the students who collected the most balls. The indices should be printed in **ascending** order.

### Scoring

| Subtask | Points | Constraints                                                                  |
|---------|--------|------------------------------------------------------------------------------|
| 1       | 14     | $n, k \leq 1000$                                                             |
| 2       | 26     | The sequence $s$ is a permutation, $s_i \neq s_j$ ( $1 \leq i < j \leq n$ ). |
| 3       | 30     | No additional constraints.                                                   |

Correctly printing the 1st line earns 50% points for the test example. The remaining 50% points for the test example are earned by correctly printing the 2nd line.



## Examples

**input**

2 1  
5 6  
2 1

**output**

6  
1

**input**

4 2  
5 5 5 5  
1 2 1 1

**output**

15  
1

**input**

4 10000000  
1 2 3 4  
2 1 4 3

**output**

4  
4

**Clarification of the first example:** In the only round of krugomet, people labeled 1 and 2 threw all their balls towards each other and "swapped" how many balls each person had. After the first round, person labeled 1 had the most balls and was holding 6 balls.



## Task Harmonija

In the city of Shumen, Bulgaria, there is a large magic tree of  $n$  connected nodes, where each node hides two magic values: red  $c_i$  and blue  $p_i$ . Each value represents the power that the node gives depending on its chosen color. Your goal is to explore the harmonious paths in the tree.



Let's consider the **shortest** path from the starting node  $A$  to the ending node  $B$  in the tree. We traverse the nodes in order and choose either blue or red color for the current node. After choosing the color of the current node, the state of the path must be harmonious, i.e. no color may "mogg" another. We say that one color "moggs" another if we have chosen it at least **three more** times than the other color. This rule must hold at **every moment** of the path for the path to be considered harmonious.

We define the value of a path as the sum of the values of all nodes on that path, where each node's value is its magic value of the color it is colored with.

Your task is to determine the maximum value of  $q$  paths given with a starting and ending node if the paths **must** be harmonic.

According to the given definition, it can be shown that there is always at least 1 harmonic path between any 2 nodes in the tree.

### Input

The first line contains the natural numbers  $n$  and  $q$  ( $1 \leq n, q \leq 10^5$ ), the number of nodes in the magic tree and the number of queries

The second line contains the  $n$  natural numbers  $c_i$  ( $-10^9 \leq c_i \leq 10^9$ ), the red values of the nodes.

The third line contains the  $n$  natural numbers  $p_i$  ( $-10^9 \leq p_i \leq 10^9$ ), the blue values of the nodes.

In the next  $n - 1$  lines, there are two natural numbers  $u$  and  $v$  ( $1 \leq u, v \leq n, u \neq v$ ), there is an edge between the cities  $u$  and  $v$ . There will be a path between any two nodes in the tree.

In the next  $q$  lines, there are two natural numbers  $u$  and  $v$  ( $1 \leq u, v \leq n$ ), the labels of the starting and ending nodes of the path.

### Output

For each query, print one number on a separate line - the highest value of the harmonic path between the starting and ending nodes of the query.

### Scoring

| Subtask | Points | Constraints                |
|---------|--------|----------------------------|
| 1       | 27     | $n, q \leq 15$             |
| 2       | 41     | $n, q \leq 1000$           |
| 3       | 19     | $q \leq 10000$             |
| 4       | 23     | No additional constraints. |



## Examples

**input**

```
4 1
10 10 10 10
-10 0 -10 0
1 2
2 3
3 4
1 4
```

**output**

30

**input**

```
5 3
-5 -4 0 -3 3
3 1 -5 0 0
3 2
1 4
3 5
1 2
2 5
1 4
5 3
```

**output**

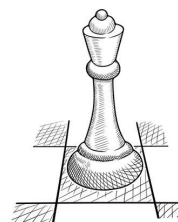
4  
3  
3

**Clarification of the first example:** If we color node 1 red, node 2 blue, and nodes 3 and 4 red, we get a harmonic path with a value of 30. There is no harmonic path with a higher value because we have to color at least one of the 4 nodes blue, and the blue values do not increase the sum in this example.



## Task Kraljica

Queen Sofia is trapped on a mysterious chessboard of  $n$  rows and  $m$  columns. The board contains her starting square (labeled "S") and her exit square (labeled "E"). Unfortunately, the path to the exit is not easy - some squares are blocked (labeled "#"), while others are free (labeled ".").



But the board also hides magical portals! Some of the numbers between 1 and 9 appear exactly **twice** on the board. If the queen lands on a square with a number, she **can** teleport to another square on the board with the same number without spending extra moves to move through the portal. The queen cannot continue moving in the same direction after passing through the portal without an additional cost of 1 move.

The queen can move in a chess-like manner - in one move she can jump to any square that is on the same diagonal, row, or column as her, but only **if there are no blocked squares between her and that square** (including the starting and ending squares of the move).

Find and print the smallest number of moves required for the queen to reach the exit square from the starting square, or print -1 if this is not possible.

### Input

The first line contains the integers  $n$  and  $m$  ( $1 \leq n, m \leq 1000$ ), the number of rows and the number of columns of the board.

Each of the next  $n$  lines contains  $m$  characters  $c_{ij}$ , the description of the board. Each character of the board will be one of S, E, a number between 1 and 9, "." or "#" (without quotes).

### Output

In the first and only line, print a single number - the smallest number of moves required for the queen to reach the exit square.

### Scoring

| Subtask | Points | Constraints                              |
|---------|--------|------------------------------------------|
| 1       | 24     | $n, m \leq 5$                            |
| 2       | 32     | There won't be any portals on the board. |
| 3       | 18     | $n, m \leq 500$                          |
| 4       | 36     | No additional constraints.               |



## Examples

**input**

```
4 4
S..1
####
1.#E
....
```

**output**

```
4
```

**input**

```
3 3
S..
.#.
..E
```

**output**

```
2
```

**input**

```
5 4
S.21
####
2##1
###.
E..#
```

**output**

```
4
```

**Clarification of the third example:** Queen Sofia will move to number 1 in the 1st row in her first move (note that there is no blocked square between S and 1, so this move is indeed possible). She will then teleport to the second number 1 in the matrix and will make 3 more moves to the exit square. With these moves, she has indeed made the fewest number of moves needed from the starting to the exit square.



## Task Zagi

Jakov and Toni, the two best players in the world, will face each other in the finals of the world championship in *krastoboj*.

Krastoboj is a game played by two players on a sequence of positive integers. The players take turns making moves, and the player who cannot make a move loses the game. At the very beginning, there is only one sequence of positive integers on the board - the initial sequence.

In one move, a player chooses **one** of the existing sequences and **one** number  $x$  that appears in that sequence. Then they delete **all** occurrences of  $x$  in the chosen sequence, thereby **splitting** the sequence into several new sequences separated at the positions where  $x$  appeared.

A sequence of numbers - the template for the final match - has been found. It is known that the initial sequence in the final will be some **contiguous** subsequence of this template sequence. You are given  $q$  scenarios. For each scenario, can you determine who the winner will be, assuming that both Toni and Jakov play optimally and that Toni makes the first move?



### Input

The first line contains two positive integers  $n$  and  $q$  ( $1 \leq n, q \leq 10^5$ ).

The second line contains a sequence of positive integers  $a_1, a_2, \dots, a_n$  ( $1 \leq a_i \leq 32$  for each  $i = 1, 2, \dots, n$ ).

The following  $q$  lines each contain two integers  $l_i$  and  $r_i$  ( $1 \leq l_i \leq r_i \leq n$ ), the boundaries of the contiguous subsequence considered in the  $i$ -th scenario.

### Output

For each query, print on a separate line either **Toni** or **Jakov**, the name of the winner in the  $i$ -th scenario.

### Scoring

| Subtask | Points | Constraints                                     |
|---------|--------|-------------------------------------------------|
| 1       | 15     | $n, q \leq 10$                                  |
| 2       | 11     | $n, q \leq 1000, a_i \leq 2$                    |
| 3       | 18     | $n, q \leq 1000$                                |
| 4       | 14     | $a_i \leq 2$                                    |
| 5       | 23     | $a_{l_i} = a_{r_i}$ for every $i = 1, \dots, q$ |
| 6       | 29     | No additional constraints.                      |



## Examples

**input**

```
6 4
1 3 2 3 1 2
1 1
2 3
2 4
1 3
```

**output**

```
Toni
Jakov
Toni
Toni
```

**input**

```
10 5
3 3 3 1 2 2 1 2 2 1
2 3
9 10
5 6
5 8
3 7
```

**output**

```
Toni
Jakov
Toni
Toni
Toni
```

**Clarification of the first example:** In the third scenario, Toni chooses  $x = 2$ , which splits the sequence into two sequences of length 1. Whichever sequence Jakov chooses next, Toni will choose the remaining one, after which Jakov will have no possible moves left.